

Boost Converter Power Stage Design

Given Parameters

- Input Voltage: $V_{\text{in}} = 12 \text{ V}$
- Output Voltage: $V_{\text{out}} = 48 \text{ V}$
- Output Power: $P_{\text{out}} = 120 \text{ W}$
- Switching Frequency: $f_s = 100 \text{ kHz}$
- Ripple Current Rate: 30%
- Ripple Voltage Rate: 1%

1. Duty Cycle (D)

$$D = 1 - \frac{V_{\text{in}}}{V_{\text{out}}} = 1 - \frac{12}{48} = \boxed{0.75 (75\%)}$$

2. Output Current (I_{out})

$$I_{\text{out}} = \frac{P_{\text{out}}}{V_{\text{out}}} = \frac{120}{48} = \boxed{2.5 \text{ A}}$$

3. Output Resistance (R_{out})

$$R_{\text{out}} = \frac{V_{\text{out}}}{I_{\text{out}}} = \frac{48}{2.5} = \boxed{19.2 \Omega}$$

4. Inductor Ripple Current (ΔI_L)

$$\text{Average Inductor Current: } I_{L_{\text{avg}}} = \frac{I_{\text{out}}}{1 - D} = \frac{2.5}{1 - 0.75} = 10 \text{ A}$$

$$\text{Ripple Current: } \Delta I_L = 0.3 \times I_{L_{\text{avg}}} = 0.3 \times 10 = \boxed{3 \text{ A}}$$

$$\text{Max/Min Current: } I_{L_{\text{max}}} = I_{L_{\text{avg}}} + \frac{\Delta I_L}{2} = \boxed{11.5 \text{ A}}$$

$$I_{L_{\text{min}}} = I_{L_{\text{avg}}} - \frac{\Delta I_L}{2} = \boxed{8.5 \text{ A}}$$

5. Output Capacitor (C)

$$\Delta V_{\text{out}} = 0.01 \times V_{\text{out}} = 0.01 \times 48 = \boxed{0.48 \text{ V}}$$

$$C = \frac{I_{\text{out}} \cdot D}{f_s \cdot \Delta V_{\text{out}}} = \frac{2.5 \times 0.75}{100,000 \times 0.48}$$
$$= \boxed{39.06 \mu\text{F} \text{ (Use } 47 \mu\text{F}, 63 \text{ V)}}$$

6. Inductor (L)

$$L = \frac{V_{\text{in}} \cdot D}{f_s \cdot \Delta I_L} = \frac{12 \times 0.75}{100,000 \times 3} = \boxed{30 \mu\text{H} \text{ (Use } 33 \mu\text{H)}}$$

7. MOSFET Selection

- Voltage Rating: $\geq 48 \text{ V}$ (Choose 60 V or 100 V)
- Current Rating: $\geq 11.5 \text{ A}$ (Choose $\geq 15 \text{ A}$)
- Example: **IRF540N** (100 V, 33 A)

8. Diode Selection

- Reverse Voltage: $\geq 48 \text{ V}$ (Choose 60 V or 100 V)
- Current Rating: $\geq 2.5 \text{ A}$ (Peak $\geq 11.5 \text{ A}$)

- Example: **STPS20H100CT** (100 V, 20 A Schottky)

Component Summary

Component	Value/Part	Specification
Inductor	33 μ H	Saturation current $\geq$ 12 A
Capacitor	47 μ F, 63 V	Low ESR
MOSFET	IRF540N	100 V, 33 A
Diode	STPS20H100CT	100 V, 20 A Schottky